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9.STATIC CODE ANALYSIS USING SONARQUBE

Java Duplicate Code and Maintainability Analysis

1. Objective

To perform static code analysis of a Java program using SonarQube, identify duplicate code and maintainability issues, and verify the analysis results before and after improving the code.

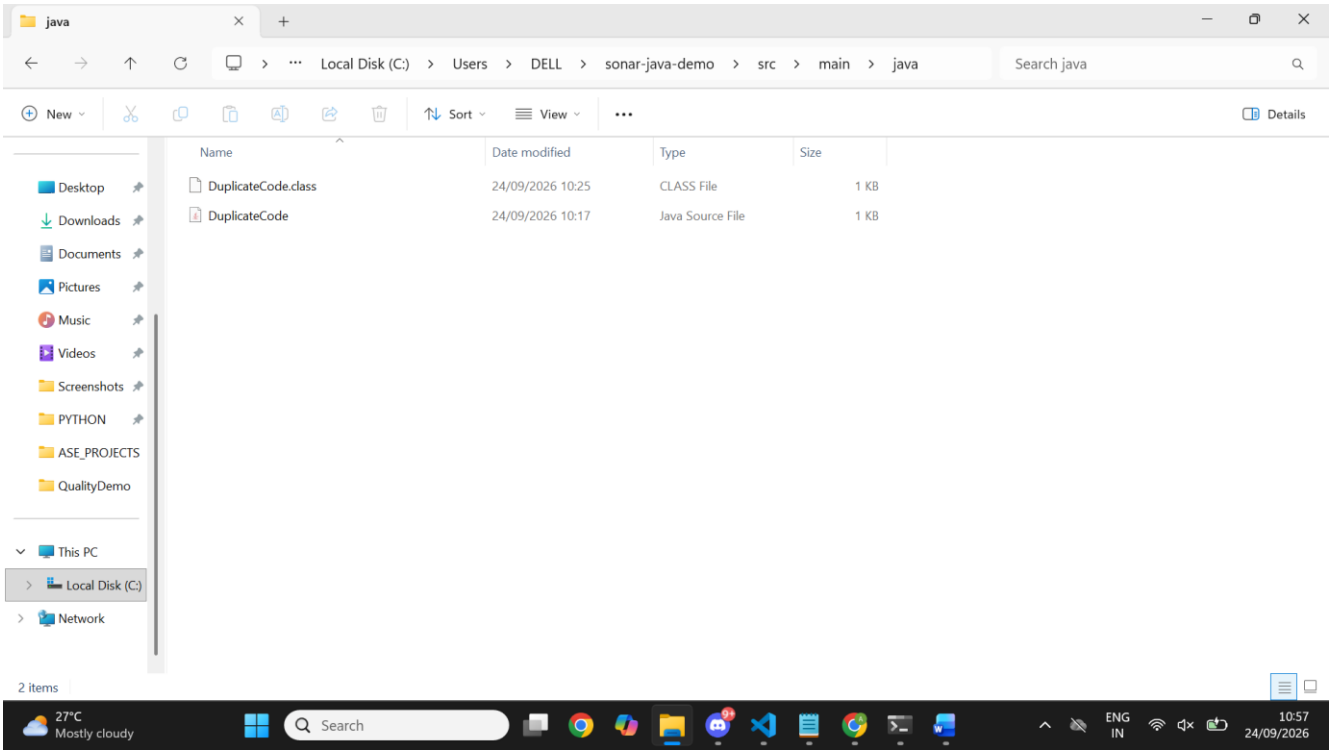
2. Software and Requirements

- Windows operating system
- Java JDK installed
- SonarQube Server installed locally
- SonarScanner CLI installed
- A Java source file containing duplicate statements
- Web browser

3. Project Folder Structure

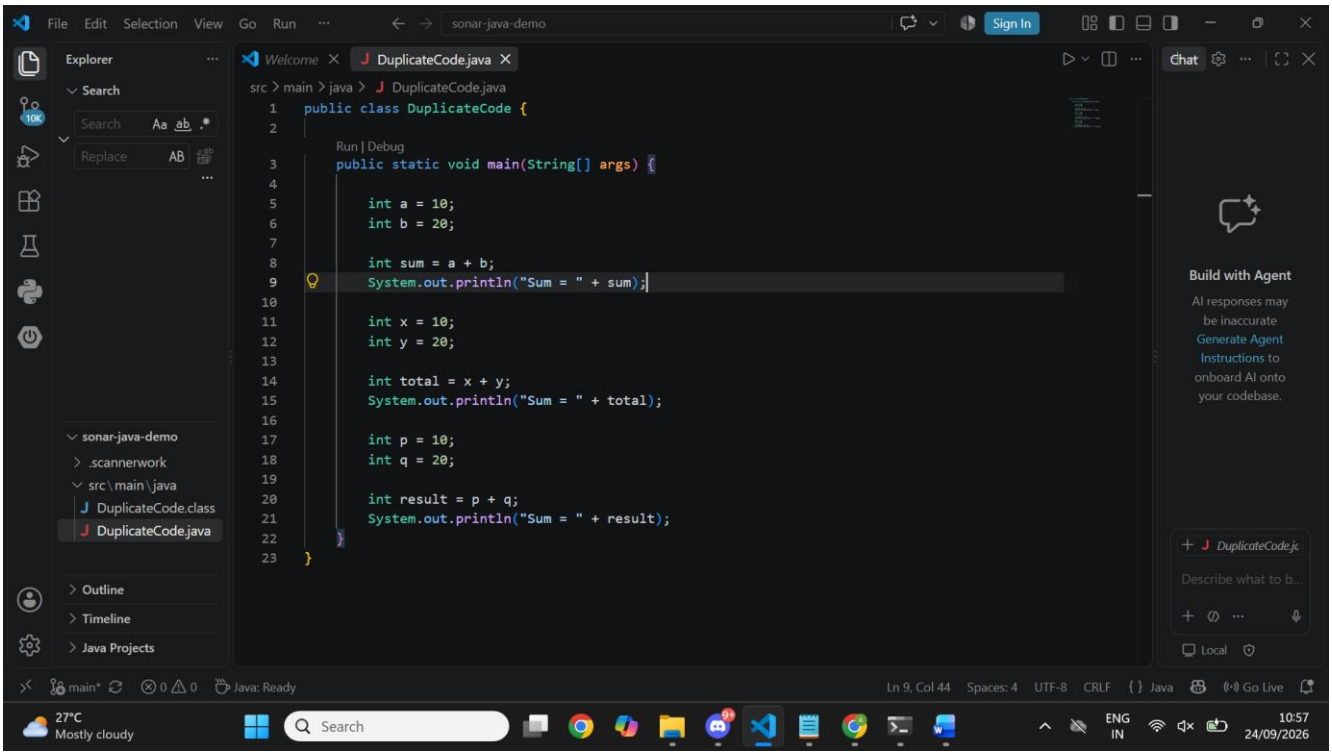
Create the following project structure:

C:\Users\DELL\sonar-java-demo\src\main\java\DuplicateCode.java



4. Create the Initial Java Program

Create a file named DuplicateCode.java and add the following code:



and Run the Java Program

Open Command Prompt and execute:

cd /d C:\Users\DELL\sonar-java-demo
javac src\main\java\DuplicateCode.java
java -cp src\main\java DuplicateCode

Expected output:

Sum = 30
Sum = 30
Sum = 30

```
C:\Users\DELL\sonar-java-demo>javac src\main\java\DuplicateCode.
java

C:\Users\DELL\sonar-java-demo>java -cp src\main\java DuplicateCo
de
Sum = 30
Sum = 30
Sum = 30

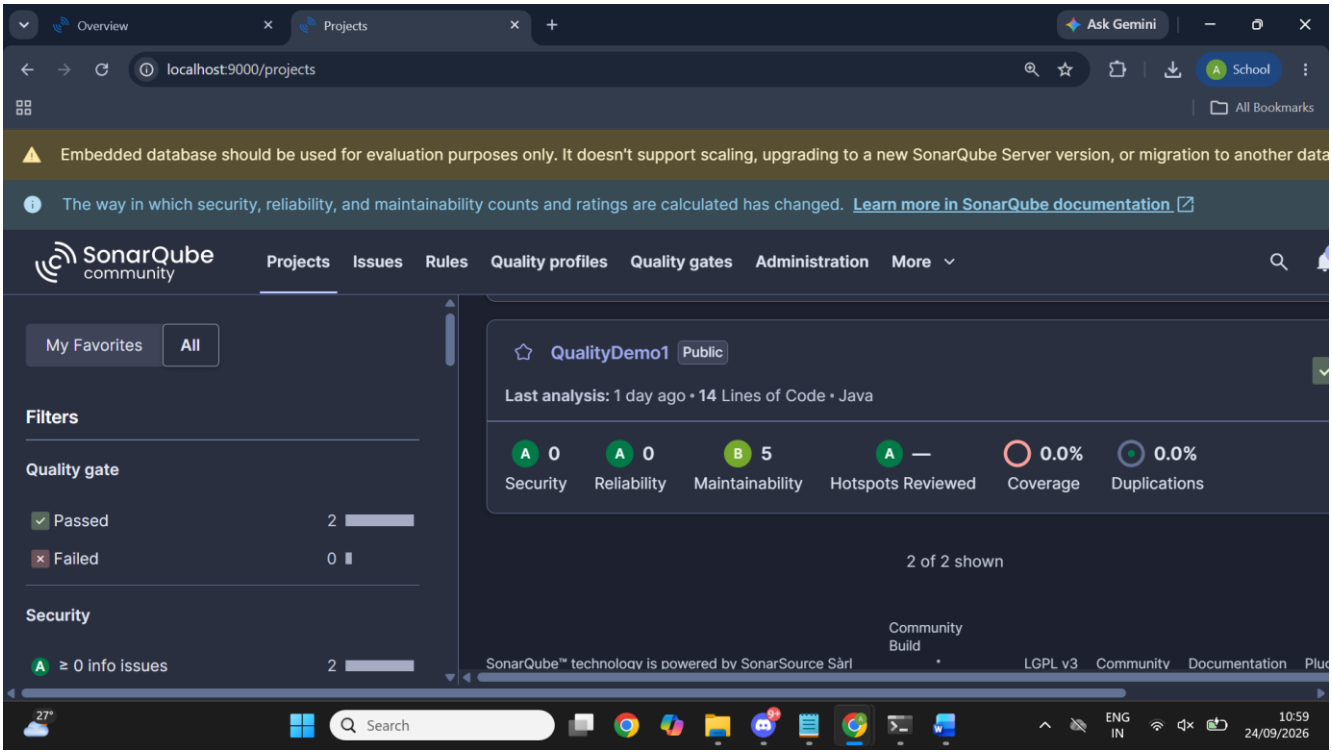
C:\Users\DELL\sonar-java-demo>
```

6. Start SonarQube Server

Open Command Prompt and run:

cd /d C:\Users\DELL\sonarqube-26.9.0.129388\bin\windows-x86-64
StartSonar.bat

After the server starts, open http://localhost:9000 in a browser and log in to SonarQube.



7. Create the SonarQube Project

Create a local project in SonarQube. Use the project name:

Java Duplicate Code Analysis

The project key used for the successful analysis is:

Java-Duplicate-Code-Analysis

For the new-code definition, the instance default (Previous version) can be retained.

8. Generate a SonarQube Analysis Token

Generate an analysis token in SonarQube and keep it private. Do not include the token in screenshots or in this report.

9. Verify SonarScanner Installation

Run:

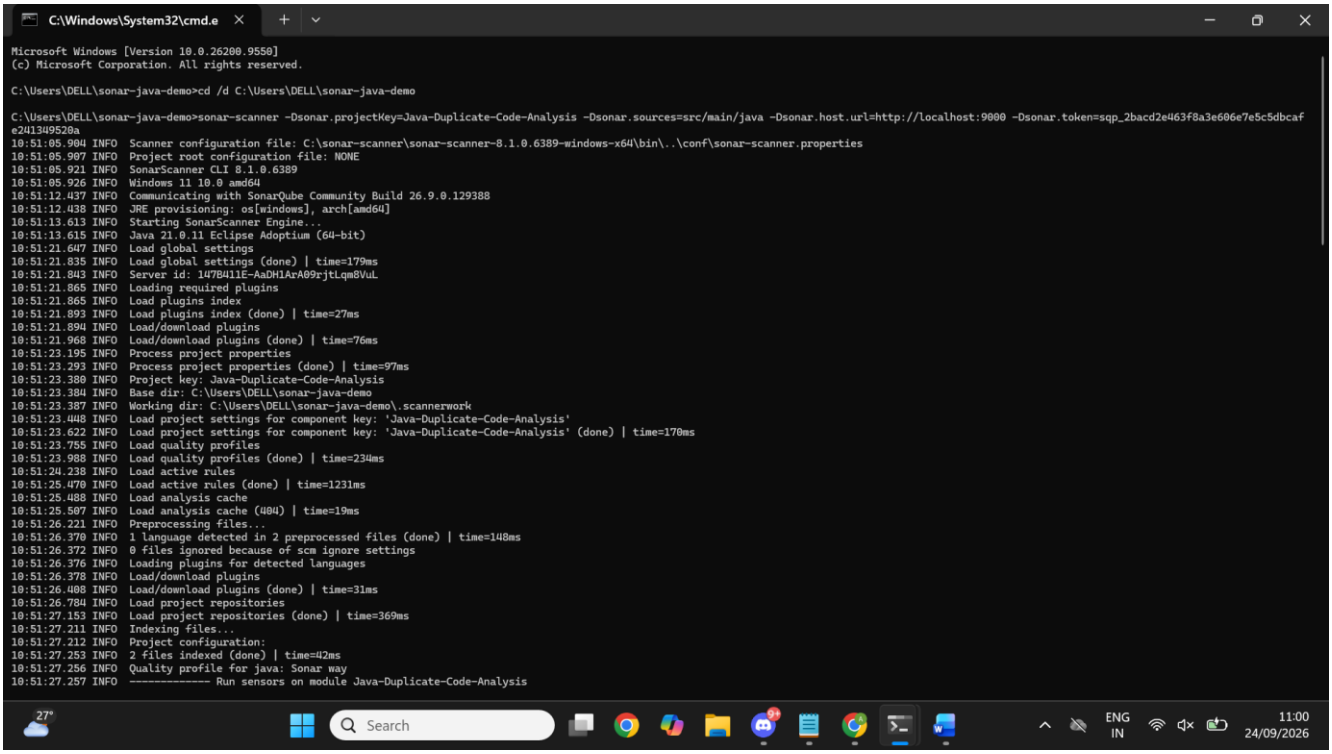
```
sonar-scanner -v
```

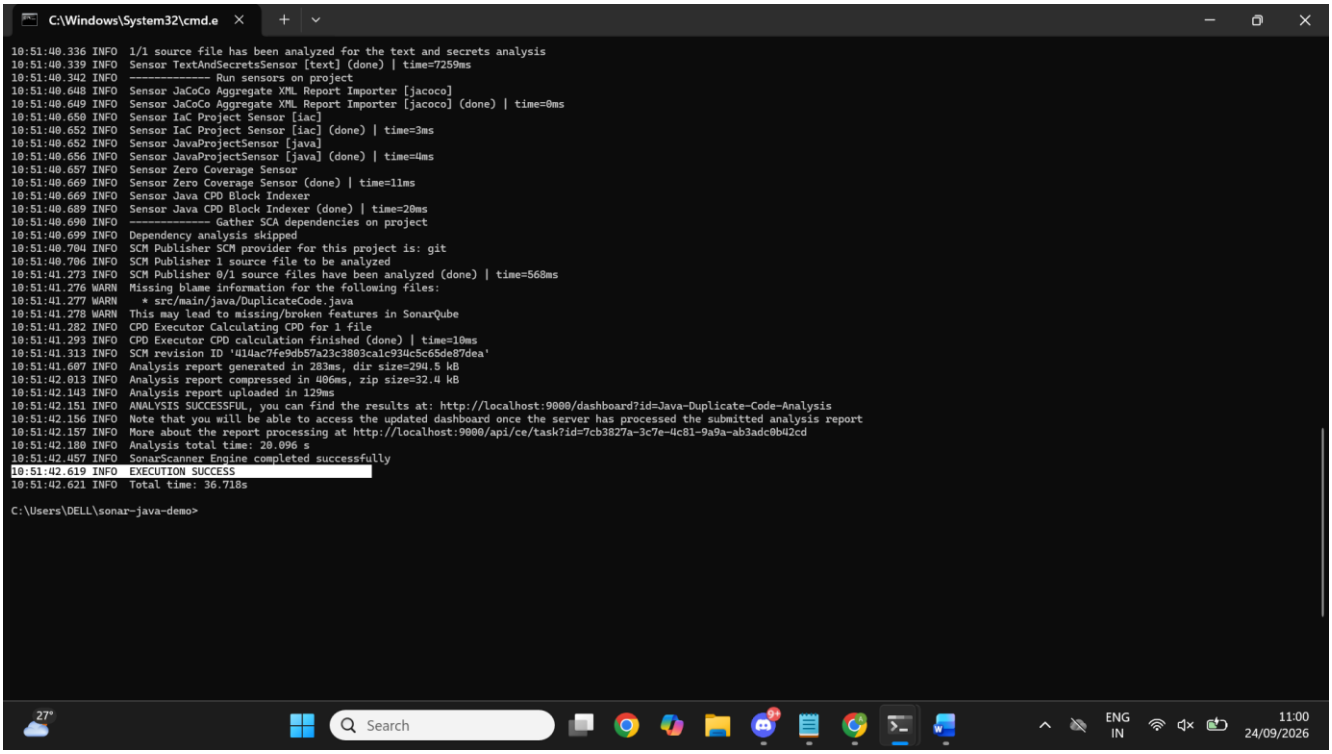
The command should display the installed SonarScanner CLI version and the operating system information.

10. Run the First SonarQube Analysis

Open Command Prompt and run the following from the project directory:

```
cd /d C:\Users\DELL\sonar-java-demo
sonar-scanner -Dsonar.projectKey=Java-Duplicate-Code-Analysis -Dsonar.sources=src/main/java -Dsonar.host.url=http://localhost:9000 -Dsonar.token=YOUR_NEW_TOKEN
```



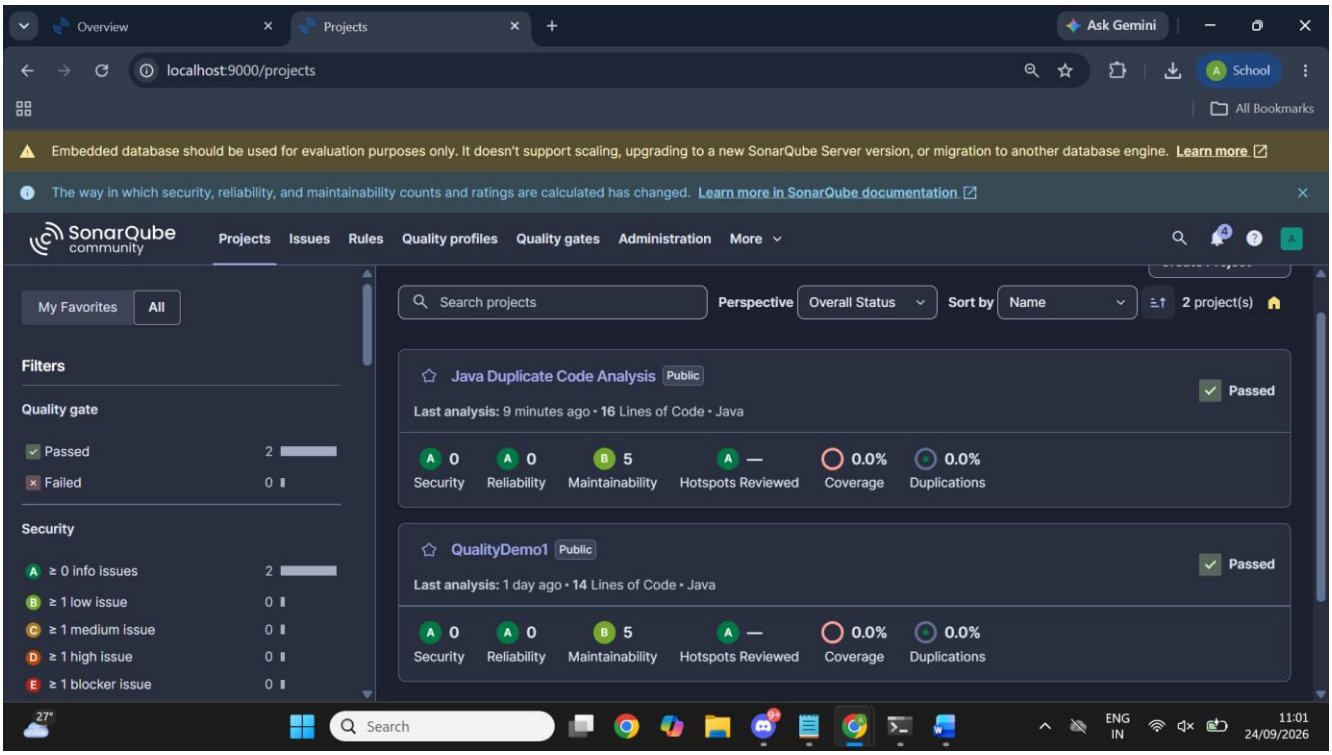


11. Open the SonarQube Dashboard

Open the project dashboard in SonarQube. The project URL used in this project is:

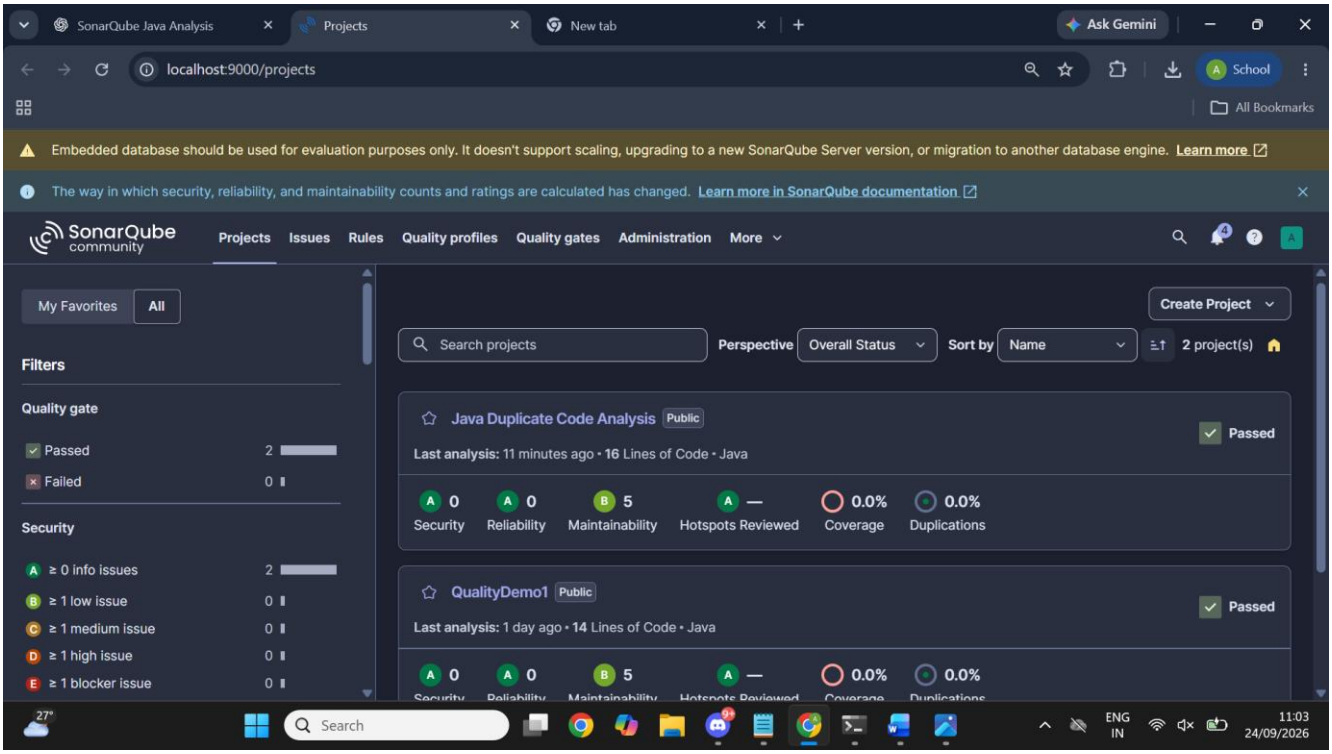
http://localhost:9000/project/dashboards/built-in/project-health?id=Java-Duplicate-Code-Analysis

The initial analysis showed 16 lines of code, Security A with 0 issues, Reliability A with 0 issues, Maintainability B with 5 issues, 0.0% coverage, and 0.0% duplications.



12. Conclusion

SonarQube can be used for static code analysis of Java applications to identify code-quality problems such as maintainability issues and code duplication. The project demonstrated the complete process of creating a Java program, running SonarQube analysis, reviewing the dashboard, identifying repeated code, and refactoring the program to improve maintainability.


SonarQube Java Analysis
Projects
New tab
Ask Gemini
localhost:9000/projects
School
All Bookmarks
Embedded database should be used for evaluation purposes only. It doesn't support scaling, upgrading to a new SonarQube Server version, or migration to another database engine. Learn more
The way in which security, reliability, and maintainability counts and ratings are calculated has changed. Learn more in SonarQube documentation
SonarQube community
Projects
Issues
Rules
Quality profiles
Quality gates
Administration
More
My Favorites
All
Filters
Quality gate
Passed 2
Failed 0
Security
≥ 0 info issues 2
≥ 1 low issue 0
≥ 1 medium issue 0
≥ 1 high issue 0
≥ 1 blocker issue 0
Create Project
Search projects
Perspective
Overall Status
Sort by
Name
2 project(s)
Java Duplicate Code Analysis
Public
Passed
Last analysis: 11 minutes ago • 16 Lines of Code • Java
0 Security
0 Reliability
5 Maintainability
— Hotspots Reviewed
0.0% Coverage
0.0% Duplications
QualityDemo1
Public
Passed
Last analysis: 1 day ago • 14 Lines of Code • Java
0
0
5
—
0.0%
0.0%
27°
Search
ENG
IN
11:03
24/09/2026

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